Phan Chi Kien

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OBJECTIVE

Third-year Computer Engineering student experienced in C/C++ firmware, Linux-based edge systems, microcontrollers, and communication protocols. Seeking an **Embedded Software internship (C/C++/Linux)** to analyze requirements, design modules, integrate and test software, document results, and debug failures.

EDUCATION

University of Information Technology – VNU-HCM (UIT) 2023 – 2027 (expected)
Bachelor of Computer Engineering GPA: 7.7/10 | 3rd Year

SKILLS

Programming: C/C++, Python, Bash, Verilog HDL
Embedded / OS: Linux (Ubuntu/Debian), bare-metal firmware, FreeRTOS fundamentals, STM32F103, ESP32, Arduino, MaixCAM
Peripherals / Protocols: GPIO, ADC, DMA, Timer/PWM, interrupts, UART, SPI, I2C, WS2812B, HD44780
Development / Test: Git, GCC/GDB, CMake, Make, Docker, OpenOCD/JTAG, ModelSim, logic analyzer

PROJECTS

FFT Audio Spectrum Analyzer & LED Matrix 2025
C/C++, STM32F103C8T6 (ARM Cortex-M3), ADC, DMA, PWM, HD44780 | GitHub

- **Embedded C/C++:** Built a bare-metal audio analyzer that processed 10 kHz ADC samples into 10 frequency bands from 200–4,000 Hz by implementing a 128-point Q15 fixed-point Cooley–Tukey FFT on STM32F103.
- **Driver Integration & Debugging:** Controlled 60 WS2812B LEDs at 800 kHz and updated a custom LCD through timer/PWM, DMA, GPIO, and state-machine modules. Diagnosed power, keypad, and timing faults with iterative tests and a logic analyzer.

Edge-based Facial Recognition Flight Information System 2026
Python, Linux, MaixCAM (RISC-V SoC), FastAPI, ONNX, Docker | GitHub

- **Linux Embedded Integration:** Designed a 3-tier offline kiosk linking a web client, FastAPI server, and MaixCAM Linux edge node. Synchronized 200 passenger profiles in under 500 ms through a 3-level cache.
- **Testing & Reliability:** Implemented and tested face matching, cache persistence, fallback behavior, and encrypted data flows on a RISC-V SoC. Achieved 15.86 ms inference latency and 0.22% FAR while documenting failure modes.

Hardware SHA-256 Cryptographic Core 2026
Verilog HDL, ModelSim, Quartus Prime, FSM, Digital Logic | GitHub

- **Low-level Design & Verification:** Designed an SHA-256 core with an FSM and 16-word message schedule. Constrained its clock to 50 MHz and verified component and top-level behavior against known vectors in ModelSim.

ACHIEVEMENTS & LEADERSHIP

- **3rd Place, IT Hackathon Solana Pragmatic 2025:** Built a working prototype under a short deadline.
- **Top 36/100 Teams, Google Hackathon Vietnam 2025:** Converted ambiguous requirements into testable behavior.
- **Lead of Design and Media, WebDev Studios (UIT):** Coordinated cross-team plans and documented public communications for club activities.